Cultural Comparisons of Beliefs and Values: Applying the Grid-Group Approach to the World Values Survey

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The goal of our research is to enrich quantitative cross-cultural studies by employing concepts of grid and group, which were first developed within anthropology and more recently deployed in a broader range of social sciences. The grid-group approach has until now not been subjected to large-scale quantitative empirical investigation, thus limiting its use in cross-cultural research. In order to carry out such an investigation, we develop individual-level indices for measuring grid and group, using data from the World Values Survey, and then investigate the collective or cohort level by taking a country or a city as a main cohort proxy and making comparisons across different cultures. We establish a number of results that suggest the usefulness of grid and group in quantitative studies: (a) grid and group variables exhibit significantly less variance within societies than between societies, (b) the characterizations of societies derived from the indices coincide with findings in the ethnographic literature, and (c) grid and group are statistically related to a society's socioeconomic conditions and geographic location. We hope our results will further stimulate the formulation of empirically testable propositions using cultural indices.

This article examines cultural differences across countries by using the grid-group framework. The grid-group framework is a well-known cultural representation framework in cultural anthropology. It is a typology of social environments created by anthropologist Mary Douglas (1970, 1982) that has been adapted, modified, and

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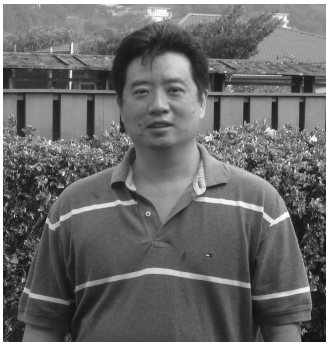

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applied over the subsequent years to develop into a major body of literature not only in anthropology but also in political science, public policy, and related social science disciplines (Chai 1997, 2001; Chai & Swedlow, 1998; Ellis & Thompson 1997; Lockhart 2001; Thompson, Ellis, & Wildavsky 1990; Wildavsky, 1983).

While the grid-group framework has been applied to culture at multiple levels of analysis (individual, regional, national) as well as to material culture, our focus in this article is on its applicability to organizing and explaining "individual culture" (i.e., attitudes such as beliefs and values). However, the relationship between culture at an individual and group level is a mutually constituting one. Individuals who share similar values generate norms and customs that are institutionalized within group practices. Likewise, these institutions act to socialize and form the individual values and beliefs of later generations. Hence, while most much of our analysis is at an individual level, it is implicit that values and beliefs are the bases for the "ways of life" that constitute culture at the aggregate level.

Sociologists Kemper and Collins (1990) conclude that the structural features underlying the two-dimensional (grid-group) model can be usefully applied not only to comparisons between classes and occupational and professional groups but also to cross-cultural comparisons between whole societies. Hitherto, the model has been used less in cross-cultural comparisons than in limited comparisons of smaller units of analysis (Caulkins, 1999).

The grid-group framework proposes that an individual's behaviors, perceptions, attitudes, beliefs, and values are shaped and regulated by cultural domains that can be labeled as group commitment and grid control. Beliefs about humans and their world characterize individuals with respect to the grid and group dimensions and spawn preferences for specific patterns of social relations. The *group* dimension represents the extent to which a culture emphasizes a merging of an individual's interests into the interests of the group as opposed to the pursuit of self-interest. The *grid* dimension represents the extent to which a culture embodies a reliance on standardized role-based rules for achieving goals as opposed to more general approaches to problem solving. While these two dimensions are deceptively simple, they also provide a systematic framework for organizing large numbers of more specific cultural features.

One important aspect of grid and group is their implications for behavior. In general, low-grid people tend to pursue novelty and do not shy away from risky behaviors, while high-grid people display routine, patterned behavior and avoid risky or unfamiliar situations. Low-group people may be more self-focused in their behavior and competitive with others. High-group people tend to see their group as more important than themselves and seek to subsume themselves via collectivist behavior. In addition to such general observations, this framework has implications for more context-specific behaviors, such as participation in the political process and the use of violent collective action (Chai & Wildavsky, 1994).

Grid-group theory surmounts some of the limitations associated with previous models of cultural variation. Most notably, it is perhaps the only general framework that has clear implications for action across the broad range of choice contexts. This feature is partly due to its origins as a general model of cultural variation rather than a case-specific description of particular societies. For this reason, it has a greater resemblance to the general cultural classifications of the sociological theories of the late 19th and early 20th centuries than it does to the models of cultural variation found in contemporary cross-cultural psychology or psychological anthropology. Indeed, its closest counterparts are Durkheim's concepts of regulation and integration (Durkheim, 1997). This general scope helps link the specifics of empirical research on culturally situated actors with the more abstract and predictive functions of social science theory (Chai & Wildavsky, 1994). Moreover, the grid-group model is rare in its complementarity with general decision-making models, such as game theory and expected utility theory.

Most previous empirical research using the grid-group model begins with ethnographic particulars and then employs grid-group theory to contextualize ethnographic findings within the range of cultural variation predicted by the grid-group model. As a result, such empirical research can neither precisely capture the differences between specific countries and cultural contexts nor help relate cultural dimensions to broader literatures on human social and economic behavior.

To address these issues via quantitative surveys and systematically analyze cultures, Gross and Rayner (1985) have presented a mathematical model for quantitative analysis by combining various survey questions into two variables: the grid score and the group score. While this effort was a major breakthrough in grid-group research, it is limited by the need to administer lengthy surveys. Furthermore, the nature of the survey items, many of which depended on relational information about subjects, presupposed a census-like coverage of a collectivity whose culture was being studied. Both characteristics make it impossible to extend the approach beyond the kinds of small communities to which it was originally applied in anthropology, much less to the extremely large aggregate units involved in cross-national comparison.

This article attempts to overcome some of these limitations by operationalizing and applying grid-group theory in a way that can make use of existing cross-national survey data sets, thus greatly expanding the realm in which it can be applied. We demonstrate this methodology through the World Values Survey (WVS), the widest-ranging and close to the longest-running survey of cultural variables in existence (Inglehart, Basañez, Diez-Medrano, Halman, & Luijkx, 2004; Inglehart, Basañez, & Moreno, 1998). The WVS originated in 1981 as an extension of the earlier-established European Values Survey and has long been associated with the political scientist Ronald Inglehart and his colleagues at the University of Michigan as well as an international network of collaborators who supervise each country study. As of the latest, fifth wave in 2007, coverage has been extended to 97 different countries.

While the WVS is not the only large-scale study that attempts to look at values on a worldwide basis, it is the most suitable for quantitative cross-cultural comparison. The Human Relations Area Files (HRAF) is even older than the WVS and is by far the world's largest compilation of ethnographic material. However, despite its extremely wide coverage and unique tagging and classification system developed by the anthropologist George Murdock (1982), it is primarily a qualitative resource that would require extensive recoding to be comparable to the WVS. While a number of data sets have been generated from the HRAF ethnographies (Barry & Schlegel, 1980; Murdock 1967), the focus of such work has been on small societies rather than countries and on customs and practices such as child rearing rather than on attitudinal variables like values and beliefs.

We start by investigating individual-level characteristics that can be attributed to cultural context. Using existing survey questions selected from WVS, two new variables representing the cultural dimensions (*grid* and *group*) for each individual in the WVS data set are generated. By aggregating the individual-level scores, we investigate cultural characteristics at the collective or cohort level by using country or a city as a main cohort proxy and make comparisons of different cultures. We show how an aggregated grid index and group index can represent various aspects of a culture, and an ordinal index can also easily be used to make comparisons between any pair of cultures. In so doing, we will demonstrate that grid-group theory serves as a useful tool for differentiating one culture from the other.

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CULTURE AND GRID-GROUP THEORY

There are many definitions for culture. In general, culture is the common set of ideas, values, attitudes, and norms that characterizes a group of people. Culture may include all aspects of a society, thus greatly impacting how we act and think in our everyday lives (Haukelid, 2008). There are also many approaches to investigate cultures. As introduced previously, the most general approach is the grid-group theory, which, from the end of 1980s, has been called "culture theory" (Douglas, 1992; Thompson et al., 1990; Wildavsky, 1987). The Douglas model proposes that an individual's behavior, perception, attitudes, beliefs, and values are shaped, regulated, and controlled by constraints that can be categorized into two domains: group commitment and grid control. Although the grid-group idea stems from Durkheim, it is not a theory of social wholes and is perfectly compatible with a focus on individual action and interaction. As Hendry (1999) says, the two categories of *grid* and *group* "were applied not to a society *per se* but to the social context of an individual, . . ." "refer to *the socio-cognitive context in which the individual finds herself*" (Hendry, 1999, p. 559 [emphasis in original]), and begin "unambiguously . . . with the individual" (Hendry,

1999, p. 562). Grid-group analysis evolved from anthropologist Mary Douglas's growing interest as to how beliefs and social environment correlate (Duval, 2006). The grid-group concept soon became an analytical framework that was applied as a specific theory of sociality (Mamadouh, 1999). According to this framework, an individual's preferences and justifications for his or her actions shape the world of social relations. Thus, everything that an individual does or wants is seen as culturally influenced.

As with any investigation into cultural theory, questions arise regarding the appropriate unit of analysis. Because we are using attitudinal surveys as our measurement tool for culture, we are conceptualizing cultural at an individual level, that is, as a set of internalized values and beliefs. Although our primary focus is not the analysis of attitudinal impact on behavior, internalized values and beliefs are readily and validly conceptualized in mathematical form, making them compatible with contemporary behavioral (reformed) rational choice models. This integration allows for the analysis of the impact of the grid-group concept on behavior in nearly any conceivable structural setting, including those that involve interactions with other individuals. That innovation is important since a collection of individual attitudes can coalesce through processes of mutual deliberation, resulting in behavioral outcomes that are quite different from what would have occurred without this kind of collective interaction.

A related point concerns the "hardness" of the boundaries between cultural groups. With individuals increasingly able to cross boundaries, facilitating the development of syncretic, cross-cultural identities in this process, how reliably are we able to place anyone into a single cultural category, such as individualist, egalitarian, hierarchical, or fatalist? The simple answer is that we cannot in any absolute sense, nor are we intending to do so, as these four categories are simply heuristics that are meant to illuminate culture patterns at an aggregate level. We submit, therefore, that cultural groups are not clearly bounded sets for which a single label can be placed but may encompass characteristics of multiple cultures that further exhibit internal variation. Neither is there an explicit or implicit assumption or any theoretical benefit to assuming that national cultures are homogeneous in the grid and group characteristics of their populations or that individuals must fall on one extreme end or another with regard to grid and group. That said, there is of course a tendency among many grid-group theorists to assume that groups will tend to migrate to one corner or another of the grid-group matrix. However, such a conclusion, even when empirically verified, rightly remains a topic of active debate, and even the most died-in-wool advocates of the four ideal type cultures recognize the need to accommodate the possibility of cultural hybridity and mobility. In short, the explanatory power of grid and group is not grounded in any assumption of homogeneity; rather, it comes from the overarching goal of understanding the nature and expression of different cultures, the values that define and challenge them internally, and, at the level of application, how such understanding can help in the exploration and prediction of variation in behavior between individuals and groups.

As noted previously, *grid* refers to "the complementary bond of constraints on social interaction" (Gross & Rayner, 1985, p. 6) and the degree to which an individual is "constrained by external rules" (Caulkins & Peters, 2002). *Group* refers to the social unit and "represents the extent to which people are restricted in thought and action by their commitment to a social unit larger than the individual" (Gross & Rayner, 1985, p. 5). These two dimensions address "two central and eternal questions of human existence: 'who am I?' and 'how should I behave?'" (Schwarz & Thompson, 1990, p. 6). According to Hendry (1999), the grid-group framework is characterized by three main ideals: classification, power, and moral order. From these ideals, Mary Douglas developed the original schematic image (Figure 1) of the grid-group framework.

In this cultural map, Douglas categorizes the four general cultural patterns of individualism, hierarchy, egalitarianism, and fatalism generated by grid and group, her two cultural dimensions. As shown in Figure 2, the diagram can be split into four quadrants that represent four types of sociocultural context.

In this figure, Caulkins (1999) draws from descriptions from Schwarz and Thompson (1990) about attitudes toward nature to form the table. According to Caulkins (1999), the "four major configurations of [the] grid-group [framework] produce four [different] political cultures."

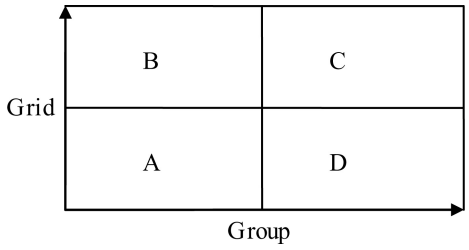

Figure 1. Douglas's cultural map: Two dimension, four types.

Source: Douglas (1978, p. 7)

High grid	B. Fatalism Apathy, risk averse Nature capricious Blame fate	C. Hierarchy Bureaucracy, decisions from above Nature perverse/tolerant Blame deviants
Low grid	A. Individualism Free exchange, competition Nature benign Blame incompetence	D. Egalitarianism/enclave Sharing, concern with moral purity and boundaries against outsiders Nature is ephemeral Blame the system
	Low group	High group

Figure 2. Grid-group analysis.

Source: Caulkins (1999)

Quadrant A displays low grid/low group, which can be seen in an entrepreneurial or individualistic environment. This type encompasses "few constraints and [has a] wide scope for forging and serving network connections freely" (Caulkins, 1999). For the sake of clarification, by grid-group typology, this type of culture is usually called "individualism." The "fatalist" environment, quadrant B, has high grid/low group and "is constrained by exterior social forces without the advantage of security within a group" (Caulkins, 1999). "Hierarchy," quadrant C, has high grid/high group and is the "classical Weberian bureaucracy with clear organization hierarchy and rule-constrained rational action" (Caulkins, 1999). Finally, quadrant D has low grid/high group and is representative of "egalitarianism," which emphasizes "group solidarity and deplore[s] extensive social differentiation," frequently using scapegoats within the social units (Caulkins, 1999).

According to Kemper and Collins (1990), dimensions of microinteraction can be seen in terms of "constraint by rules" (grid) and "incorporation into a bounded social unit" (group). Grid refers to the constraints created by an ordered structure and includes a high degree of hierarchy and ranking within the group itself. On the other hand, group is intended to show the role of group pressure on a person's ego and places a great deal of "emphasis on [the] boundaries between the society and outsiders" (Kemper & Collins, 1990, p. 47).

Many researchers have tried to find weaknesses of the grid-group framework (Beidelman, 1993; Caulkins & Peters, 2002; Hofstede & McCrae, 2004). For example, Beidelman (1993) argues that Douglas's theory that people see their opportunities in life in terms of the cultural modes of their society in which they live is not accompanied by sufficient detail regarding the basis or validity of this conclusion, nor does it explain how individual societies "work." From a different standpoint, Hofstede and McCrae (2004) argue that although many multidimensional theoretical models have been proposed—the grid-group theory included—such models represent subjective reflective attempts to order a complex reality. They are not clear about the levels of analysis at which they are supposed to apply and not to apply (i.e., society, social category, group, or individual).

Despite such criticisms, supporters of the framework have forcefully rebutted these claims. For example, according to Kemper and Collins (1990), the structural features underlying the two-dimensional grid-group model can be usefully applied not only to cross-cultural comparisons between societies but also to comparisons among classes, occupations, and professional groups. In this sense, the grid-group framework is more than a cultural taxonomy. Its two dimensions have been shown to have implications for action across a wide range of environments. Based on ethnography derived from nonindustrial societies, the model has by now been applied to a variety of institutions, themes, and areas (e.g., a review of the literature lists some 100 academic publications; see Thompson et al., 1990), such as Chinese medieval history, theater, geology and mathematical sciences (Douglas, 1982), ecology (Douglas & Wildavsky 1983), occupational crime (Mars, 1982), industrial safety (Gross & Rayner, 1985), risk behavior (Douglas, 1992), and information processing (Thompson & Wildavsky, 1986). More recently, Caulkins (2009) demonstrates that the grid-group concept continues to be applied by an interdisciplinary cadre of scholars to help explore environmentalist views, perceptions and dangers of risk, and a wide range of subcultures (e.g., public administration, religious communities, and high-technology firms).

Unlike other cultural taxonomies, grid-group theory's practical implications can be specified in a very straightforward manner, following from the fact that the focus here is on the observable behavior of the decision maker, which represents culture as preferred values and beliefs, manifesting as behavior. These preferred values and beliefs, in interaction with the structural circumstances within which an individual is situated, generate purposive behavior that often is predictable from its cultural antecedents regardless of the specific circumstances being investigated (Chai & Wildavsky, 1994, 1998). For instance, low groupness generates self-interested preferences, while high groupness generates preferences that are interdependent, including what might be designated as positive (altruism) and negative (antagonism) other-regarding preferences. Likewise, low groupness implies little differentiation in the treatment of others within or without an individual's bounded collectivity (universalism), while high groupness implies that the boundaries have significant implications

for behavior (particularism). In contrast, high gridness implies a rigid adherence to social norms, except when two such norms lead to contradictory implications for behavior. Such adherence is sustained by beliefs that ideologize norms as optimal rules for obtaining one's intrinsic goals or internalize norm-following as a goal in and of itself (Wildavsky, 1991). Low gridness, on the other hand, implies a desire for nonconformity and a belief that nonconformist behavior will be conducive to individual success. As may be clear, then, grid-group cultural theory could be and has been applied at different levels of analysis, such as how an individual fulfills social roles when they are and are not in accord with their own idealized preferences. At the same time, grid-group theory can also be applied to group-level systems, from villages to cities and from countries to world (see Mamadouh, 1999). Grid-group theory operates at multiple levels because it is conceptualized in such a way to be generally compatible with both action models, which show how individual values and beliefs can generate group institutional practices, as well as cultural change models, which further illustrate how such institutional practices may become the socializing agent for the values and beliefs of other group members.

GRID-GROUP THEORY AND THE WVS

Although there are many large data sets that can be used to investigate cultural and value differences, the most often used is undoubtedly the World Values Survey (WVS). The WVS first emerged out of the European Values Study in 1981, when the methods of a successful European study were extended to 14 countries outside Europe (European Values Study Group and World Values Survey Association, 2006). The 1981 study nevertheless covered only 22 countries worldwide. The survey was repeated after an interval of about 10 years in what came to be in effect its "second wave." Further waves followed the second wave at intervals of approximately 5 years. WVS also grew to include 42 countries in the second wave, 54 countries in the third wave, and 62 countries in the fourth wave.

WVS data are obtained through detailed questionnaires in face-to-face interviews; questionnaires from the most recent waves consist of about 250 questions. In each country, the questionnaires are administered to about 1,000 to 3,500 interviewees, with an average in the fourth wave of about 1,330 interviews per country and a worldwide total of about 92,000 interviews.

WVS questionnaires have changed over the years (e.g., the questions used in waves 3 and wave 4 differed by about 30%). Since we want to investigate the grid-group cultural dimensions for both wave 3 and wave 4,[1] we have retained the questions that were common to both of these waves and eliminated those that differed, a process that left about 148 questions[2] that were asked in both surveys. Moreover, because the WVS is not designed explicitly for measuring grid and group, many of the 148 questions retrained were not pertinent to either dimension, so additional items were also deleted. For example, political questions regarding specific political parties and political campaign–related political issues from the WVS were eliminated since they are pertinent only to a specific geographic location and political system. In contrast, general questions of a political nature, such as one regarding the proper role of government in the economy, were not excluded. Finally, while questions about happiness and life satisfaction could conceivably be correlated with certain of the grid or group dimensions, such items presumably tapped emotional states in a manner that may be different from items that represented seemingly straightforward beliefs and values about the larger world. Thus, for purposes of the present study, these items also were excluded.

The net effect of the previously mentioned review process through a grid-group lens meant that 59 questions were retained. From these, the list was winnowed further to 11 grid questions and 11 group questions, chosen because the variations across countries for these questions were higher than the rest of the questions. The intended focus of our studies, like most large-scale statistical comparative studies of culture, was to engage in comprehensive cross-national research to uncover cultural differences and variations across countries. Furthermore, grid-group theory is more often than not applied to explain cultural differences at the national level. Hence, questions that tended to show wide variation, particularly at the national level, met criteria of face validity better than those that displayed little variation.

We tried to cover as many social dimensions as possible in these 22 questions, which included the following questions, grouped by theme. Table 1 categorizes the questions within the grid-group dimension. Among them, 11 questions are assumed to reflect peoples' grid characteristics, and 11 questions are assumed to reflect peoples' group characteristics.

As shown in the table, we classified the grid-group properties for each individual question. We considered the following opinions to be indicative of a high "grid" score: (a) religion is very important, (b) men should have more rights than women, (c) young people should get more job

TABLE 1. Grid-Group Categories in Our Survey

Variable	Grid	High	Low
Grid1	Religion	Important	Not important
Grid2	Job men/women	Men more rights	Not agree
Grid3	Job old/young	Young more rights	Not agree
Grid4	Follow instruction	Yes	Not necessary
Grid5	Having children	Yes	Not necessary
Grid6	Respect authority	Yes	No
Grid7	Religion (God)	God important	Not important
Grid8	Homosexuality	Never justifiable	Justifiable
Grid9	Prostitution	Never justifiable	Justifiable
Grid10	Abortion	Never justifiable	Justifiable
Grid11	Divorce	Never justifiable	Justifiable
	Group	High	Low
Group1	Family	Important	Not important
Group2	Friends	Important	Not important
Group3	Parents	Must love and respect	Not have to
Group4	Trust people	Most can be trusted	Have to be careful
Group5	Fair pay	Not fair	Fair
Group6	Business	Employees do more	Owner do more
Group7	Importance of money	Less emphasis	More emphasis
Group8	Importance of work	Less emphasis	More emphasis
Group9	Importance of technology	Less emphasis	More emphasis
Group10	Business private/government	Government	Private
Group11	Responsibility personal/government	Government	Personal

opportunities, (d) people have to follow/obey instructions, (e) having children is quite necessary, (f) people should respect authority, (g) God is important in your life, and (h–k) homosexuality, prostitution, abortion, and divorce are never justifiable. People who belong to extreme low grid dimension will have inverse opinions on these topics. These questions are grid questions because they represent the individuation of members of the organization and the degree to which their choices are constrained by imposed rules and role descriptions. High grid refers to a social environment where individual interactions are regulated and autonomy is minimal.

Similarly, we considered the following opinions to be indicative of a high "group" score: (a) family are very important; (b) friends are very important; (c) people should respect their own parents; (d) people should trust each other; (e) efficient workers should not be paid more; (f) employees should involve more in business management; (g–i) our way of life in the future should put less emphasis on money, work, and technological development; (j) government should run business; and (k) government should take more responsibility. These questions are chosen to be group questions because they represent the value people place on collective relationships and the commitment they have to the larger social unit.

We then can determine each individual's grid-group characteristics by examining their responses to the questions used to operationalize each of the previously mentioned indicators. We assumed that people who have similar cultural backgrounds tend to share similar grid-group opinions. If the majority population of a country has a high-grid characteristic, then the country is called a high-grid country. And if most people of a country have a high-group characteristic, then the country is called a high-group country. We predict that countries that have similar cultural background will share similar grid-group characteristics. It is important to note that most countries have complex ethnic demographics and that the culture between diverse ethnicities and religions may

vary from each other. Therefore, in our study, country is only a proxy of culture because of the data restriction.

Table 2 provides the basic statistics of different variables that we chose from the WVS. The next step was to generate the aggregated index in an appropriate way. We generated our index based on equal weights because, from the questions we chose from the WVS, there was no theoretically informed a priori reason to assign one variable greater importance than another in determining gridness or groupness. Likewise, the use of principal components or factor analysis is one that is not readily compatible with deductive theories like grid-group theory. The resulting factors are designed to explain the maximum amount of joint variation within the data, not to maximize predictive power over social practices. Deriving coefficients from regression analysis has in some way the opposite problems—it does not generate factors, and while it is possible to optimize predictive power over a single data set with a selected dependent variable, there is no way to optimize for the general case. Even weighting is the most parsimonious approach and in accordance with Occam's razor avoids introducing new entities (weight coefficients) without clear justification. Moreover, even weighting is used in a number of highly reputable social indices. For example, the Human Development Index[3] by the United Nations Development Program and the Political Rights Index[4] by Freedom House use equal weights to generate their indices.

We first normalized the answers to each question onto a scale of 0 to 1 in order to avoid the inconsistent arrangement of answers for different questions in the WVS.[5] After that,

TABLE 2. Basic Statistics for the Grid-Group Component Variables

		Wave 3				Wave 4			
Variable	Question	Mean	*SD*	Rank by *SD*	*F* Value	Mean	*SD*	Rank by *SD*	*F* Value
Grid1	Religion	0.59	0.21	5	408.7	0.75	0.24	4	768.9
Grid2	Job men/women	0.44	0.17	10	107.7	0.50	0.21	6	2263
Grid3	Job old/young	0.55	0.18	9	125.6	0.74	0.22	5	174.6
Grid4	Follow instruction	0.49	0.19	8	288.6	0.56	0.19	9	592.7
Grid5	Having children	0.54	0.23	3	724.4	0.65	0.24	3	3853
Grid6	Respect authority	0.65	0.19	7	32.35	0.72	0.21	8	205.8
Grid7	Religion	0.61	0.24	1	12.97	0.80	0.21	7	286.9
Grid8	Homosexuality	0.65	0.23	2	54.35	0.74	0.27	2	803.5
Grid9	Prostitution	0.71	0.21	4	272.9	0.78	0.28	1	508.4
Grid10	Abortion	0.59	0.20	6	23.9	0.77	0.18	10	262.2
Grid11	Divorce	0.49	0.16	11	107.7	0.62	0.18	11	238.6
Group1	Family	0.93	0.13	5	874.0	0.95	0.16	3	18.77
Group2	Friends	0.74	0.13	4	718.4	0.74	0.14	5	36.32
Group3	Parents	0.79	0.16	1	47.8	0.85	0.21	1	262.4
Group4	Trust people	0.25	0.13	7	597.6	0.25	0.13	7	36.1
Group5	Fair pay	0.16	0.11	9	238.4	0.17	0.11	9	1001
Group6	Business	0.31	0.12	8	349.7	0.27	0.11	10	269.2
Group7	Importance of money	0.61	0.16	2	362.3	0.60	0.16	2	1221
Group8	Importance of work	0.29	0.13	6	202.6	0.29	0.15	4	889.8
Group9	Importance of technology	0.19	0.11	11	667.1	0.15	0.07	11	815.2
Group10	Business private/government	0.42	0.11	10	244.7	0.41	0.14	6	149.6
Group11	Responsibility personal/ government	0.57	0.11	8	169.1	0.56	0.15	3	25.9

Note. All the between-country *F* values are significant at $p < .01$.

the individual-level grid and group scores were generated by aggregating the various questions. We then calculated the country average scores from the individual scores. According to our one-way analysis of variance (ANOVA), for all questions, the *F* values show that the between-country variance is greater than the within-country variance. These results are shown as *F* values in Table 2, where the significant *F* values show that the means of these countries are not equal. In this table, we have also listed the average score and standard deviation based on the country average value, establishing a rank.

The importance of between-country differences being higher than within-country is highlighted by the fact that countries are almost universally used as the cultural unit of analysis in fields as diverse as cross-cultural psychology (Berry, Poortinga, Segall, & Dasen, 2002), cross-cultural communication (Kim, 2002), cross-cultural management (Smith, Peterson, & Thomas, 2008), and comparative studies of development (Peet, 2009). While there is some debate in these fields over the importance of subcountry group identities, pan-country regional identities, and even globalized cosmopolitanism, countries retain their hold as the "natural" units via which cultural comparisons should take place. As long as this is the case, general cultural constructs should provide sufficiently large variation between countries, even when normalized for variation within countries.

Looking at the rank ordering of the standard deviations of the scale items, the differences of opinions between countries regarding homosexuality and prostitution are large and the differences of opinions between countries regarding abortion and divorce small for both wave 3 and wave 4. Even though the opinions regarding homosexuality and prostitution vary more widely than most of the other "grid" items, the overall mean values of those two value items were still higher than the rest. The high mean value of opinions regarding homosexuality and prostitution indicates that, on average, most people do not believe that such behaviors are appropriate. People's opinions regarding children also exhibit large differences between countries. The variance of opinions regarding religion, however, is larger in wave 4 than in wave 3. This may be because the countries included in waves 3 and 4 are quite different; wave 4 included more countries that are predominantly Muslim. As a result, people's opinions on religion also seem to have changed substantially between waves 3 and 4.

The standard deviations for the group questions tend to be smaller than those for the grid questions. Among the group questions, opinions regarding the appropriate place of money and respecting parents are the two questions that have the least variance. The average value for importance of family is also very high, indicating that people in the majority of countries consider family very important. The mean values of the opinions on fair payment ("Efficient workers should not be paid more") and the importance of technology are very low, suggesting that, regardless of country, most people consider more efficient work to deserve more payment. Since we are especially interested in the differences between countries, the results based on country average scores are presented in the next section.

RESULTS OF BETWEEN-COUNTRY GRID-GROUP ANALYSES

Results of Wave 3 of the WVS

We investigated two waves of the WVS. Wave 3 included 55 countries. The average of the grid score for all 55 countries was 0.629. Nigeria (0.83) had the highest score and Sweden (0.34) the lowest. The average of the group score was 0.516. Pakistan (0.604) had the highest group score and Bangladesh (0.409) the lowest. It is noteworthy that although Bangladesh and Pakistan are both South Asian Muslim countries and historically connected with each other, their degree and nature of groupness differed. For example, it seems that more people in Bangladesh than in Pakistan think that government should run businesses.

A noteworthy first observation from the results of wave 3, as shown in Figure 3, is the strong difference of grid and group characteristics between developed countries and developing countries. Most developed countries have lower grid values than developing countries.

Second, the geographical covariance is also very strong. For example, Latvia and Estonia, which are geographically adjacent as well as newly independent from the former Soviet Union, both belong to the low-grid and low-group category. Similarly, Scandinavian countries such as Norway, Finland, and Sweden in Figure 3 are close to each other, and all belong to the low-grid and high-group category, implying that these Scandinavian countries' cultures are relatively egalitarian. Evidence from Spain and the cities belonging to Spain are also very close to each other, belonging to the low-grid and high-group category. The Latin American countries such as Peru, Puerto Rico, Brazil, the Dominican Republic, and Mexico also have

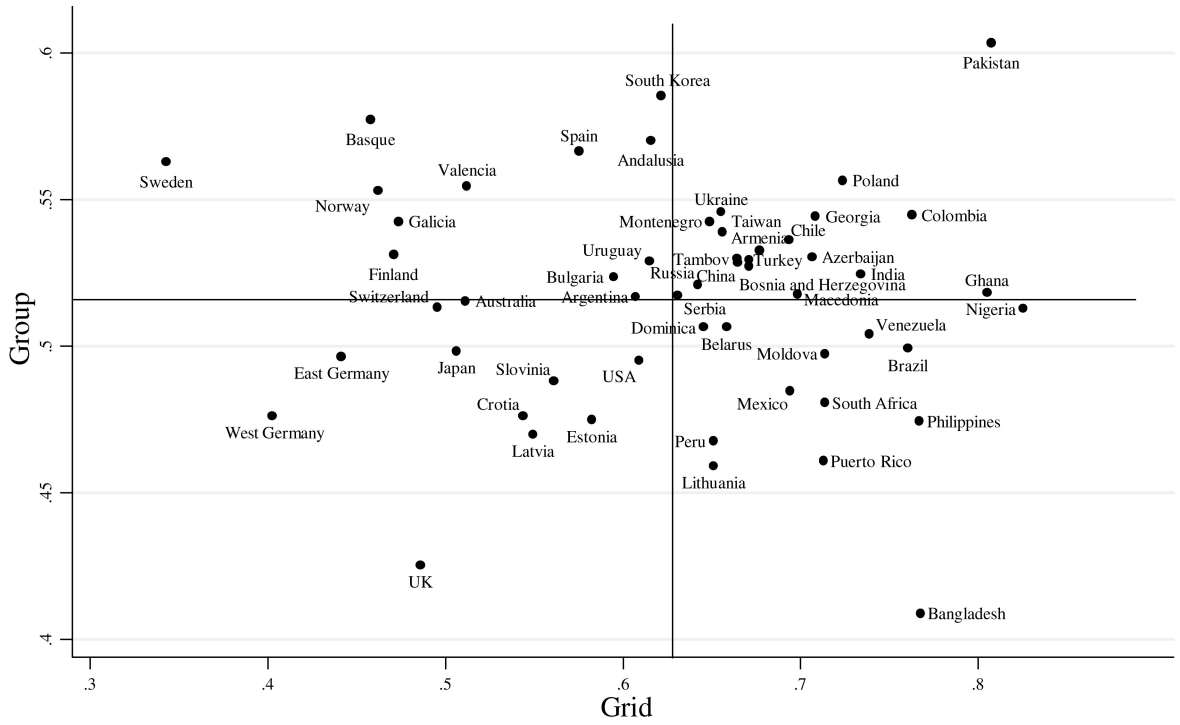

Figure 3. Relative grid-group scores in wave 3 of the WVS.

similar groupness and gridness and belong to high-grid and low-group category (e.g., "fatalism" societies).

Third, countries with a single ethnicity generally tend to have higher groupness. For example, people from newly formed nations with strong ethnic groups, such as Armenia, Georgia, Azerbaijan, and Bosnia and Herzegovina, which have very strong ethnic identifications, tend to have high group scores, which is consistent with the notion of groupness as tiedness to a collectivity. In China and Taiwan, the majority of the population is Han ethnicity, placing both countries in the high-group category. Pakistan, a Muslim country, shows extremely high grid and high group scores. As a heterogeneous country with various ethnicities, the United States belongs to the low-grid and low-group category.

It is interesting to note that although Germany and Japan are commonly considered to be hierarchical, group-oriented cultures, our results show that, when compared with other countries in wave 3 of the WVS, Germany and Japan have relatively low grid and low group scores. This finding is also reflected in the results for wave 4 of the WVS as we discuss next. In considering these results, we found that the main reasons for the low score in both Japan and Germany are two questions concerning the role of government in business and society. Germans as well as Japanese strongly disagree both that government should run business and that government should take more responsibility to provide for all members of the society. They hold these opinions much more strongly than do people from most other developed countries. The recent history of these two societies and economies may help us understand this observation. It is well known that both Japan and Germany were government-directed economies and were considered highly successful from the 1960s to the 1980s. However, since the late 1980s, both countries have undergone major economic recessions, and their government-interventionist mode of functioning has been criticized because some people argue that such a mode creates lower economic efficiency than a "free" economy. Therefore, although speculative, considering the period in which surveys were carried out for the third wave of the WVS, it is not surprising that people from Japan and Germany did not approve of "big government." In the case of gridness, Germany and Japan had lower scores because many Germans and Japanese do not agree that God is important and because many Germans and Japanese consider homosexuality, abortion, and divorce to be acceptable. This leads to our fourth observation, that gridness and groupness are affected and changed according to a country's socioeconomic and other historical context. In other words, gridness and groupness are not static but dynamic attributes of societies.

Results of Wave 4 of the WVS

The data in wave 4 include 40 cohorts. The average grid score is 0.75, dramatically higher than before, because wave 4 includes many Muslim countries, many of which typically have much higher grid scores. Bangladesh (0.924) had the highest grid score and Japan (0.523) the lowest. The average group score is 0.518. Kyrgyzstan (0.737) has the highest group score. Uganda (0.437) had the lowest group score. It is not surprising considering Uganda's political situation and complex society composition (Figure 4). Uganda has more than 14 ethnicities (Baganda, the largest ethnicity, represents only 16.7% of the population), and each ethnicity speaks a different language (Figure 4).

Some countries, such as China, were in the high-grid, high-group quadrant when classified using wave 3 data but slipped diagonally into the low-grid, low-group quadrant when using wave 4 data. That is because the wave 4 data include many more predominantly Muslim countries and their high group and grid scores drive the average scores considerably higher than those of wave 3. For the purpose of comparison, when we add the average grid score in wave 3 by using the dash line, we found that China's position is almost the same as it was in wave 3.

Results from the WVS, by using the grid-group dimensions, exhibit a predictable difference between predominantly Muslim and non-Muslim cultures. Our results show that predominantly Muslim countries tend to have higher gridness and groupness than other countries. While we should be cautious about interpreting this in terms of popular stereotypes about so-called Western and Muslim cultures, what it does show is that people in predominantly Muslim cultures tend more toward norm-governed behavior (gridness) and collective identity (groupness).

With such an observation, we investigated the relationship between Muslim culture, and the grid-group score and the results are presented in Table 3. The variables of grid-group are from WVS wave 4, and the variable of Muslim is the percentage of Muslim people in the whole population of a country. The data of Muslim percentage in 1999 for each country are obtained from Fearon and Laitin (2003). The correlation between grid score and Muslim percentage is positive and statistically significant at the 5% level. The correlation between group score and Muslim percentage is not statistically significant but also is positive. These results indicate that the difference between Muslim culture and other cultures appeared more in the grid dimension.

In Figures 5 and 6, we display the 23 countries that are common to both wave 3 and wave 4 of the WVS. We found that scores for most countries are consistent across the two surveys. It is interesting that, although the differences between the two surveys are not large, grid as well as group

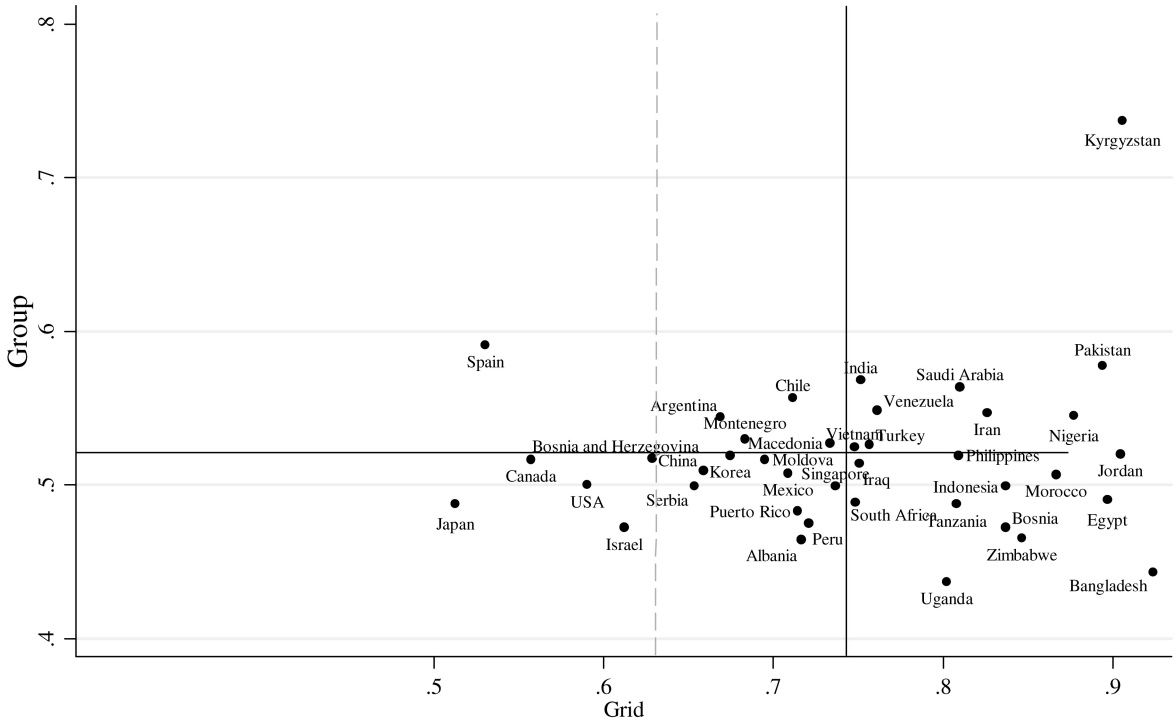

Figure 4. Relative grid-group scores in wave 4 of the WVS.

TABLE 3. Correlation Between Grid/Group Score and Muslim Culture

	Muslim Population/Whole Population × 100%
Grid score	.628* ($p < .0001$)
Group score	.237 ($p < .1841$)

scores are somewhat higher in wave 4 than in wave 3 for the majority of countries in Figures 5 and 6. From the 23 countries, only five countries have lower grid scores in wave 4 than in wave 3: the United States, Moldova, China, Bosnia and Herzegovina, and Spain. Seven countries—Japan, Serbia, China, Bosnia and Herzegovina, Montenegro, South Korea, and Pakistan—have lower group scores in wave 4 than wave 3. However, when we consider the actual values, generally the differences in grid and group scores in waves 3 and 4 were not statistically significant. For example, the grid and group scores for China in 2000 were 0.659 and 0.510, respectively, which are statistically not different from the scores in 1995 (0.665 and 0.529, respectively).

To test the sensitivity of the grid and group scores by using WVS questions other than our final set of 22 questions, we also replaced some questions with other candidate questions. For instance, we replaced "How important is God in your life?" with other religion questions such as "Do you believe in God?" and "Do you feel comfort and strength from religion?" We also replaced the question on "Justifying homosexuality and abortion" with "Cheating on taxes and bribes are never justifiable." Results indicated that scores were not significantly different when using the replaced questions, and the rank ordering of countries did not change. In other words, the alternative operationalization that we created by using the changed set of questions did not yield statistically significant differences in country scores. Based on the previous analyses, we found that the grid-group framework is a useful tool for cross-country and cross-cultural comparisons.

CONCLUSION

This article carried out cross-cultural comparisons based on grid-group theory by using data from the WVS. Overall, through the indices, we found that the grid-group model is a useful tool for cross-cultural comparisons of beliefs and values. We arrived at several important findings.

First, grid and group constitute stable social configurations that are associated with distinctive cultural values for specific countries. We have shown that these cultural value sets can be measured at the individual level through surveys and can be aggregated as well at the group level. One-way ANOVA tests indicate that there is little variability within each country but significant differences between countries. Hence, using the aggregated grid and group score would appear to capture the culture differences across countries.

Second, countries with similar socioeconomic backgrounds tend to have similar cultural characteristics. For example, developed countries have relatively less grid and group than developing countries.

Third, countries having close geographical locations also appear to have similar cultural characteristics.

Fourth and finally, ethnicity and religion play a very important role in the variation of grid and group characteristics (e.g., as we found via similar grid and group characteristics for single, newly independent countries and for Muslim countries).

We have shown that the best-known cultural taxonomy from cultural anthropology—heretofore confined largely to qualitative ethnographic analysis—can be productively applied to quantitative data in order to help make sense of differences and similarities among cultures. Through the WVS, we were able to generate culture variables that can be applied to empirical research for studying who holds which "core values" and why. Furthermore, we have demonstrated a relatively straightforward and accessible methodology for deriving grid and group from existing survey data; in so doing, we are able to identify patterns and clusters of cultural types that correspond to country, region, and civilization. Importantly, these cultural types correspond to what one would expect from the ethnographic literature.

These promising results suggest further areas of analysis. For example, efforts could be made to examine the conceptual and empirical properties of grid and group in relation to the concepts found in other quantitative cultural taxonomies, such as Hofstede's (1980, 1991) five-factor model. Specifically, it might be possible to investigate corporate culture through the grid-group variable as a comparison to the Hofstede's variables. At the same time, unlike some of Hofstede's variables, grid-group

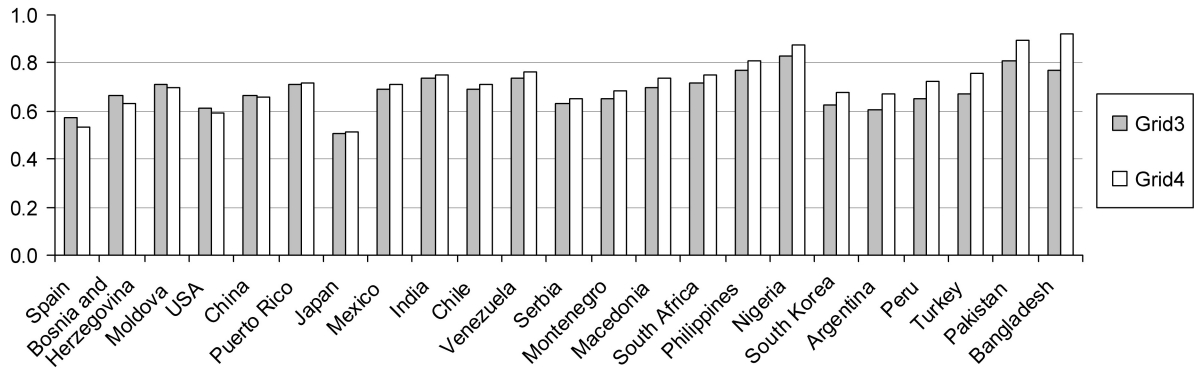

Figure 5. Comparison of grid score between waves 3 and 4 of the WVS.

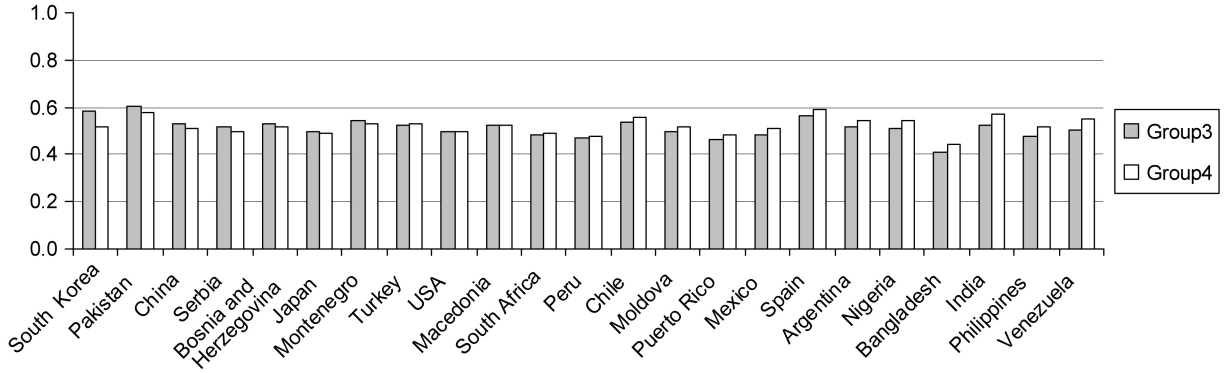

Figure 6. Comparison of group score between waves 3 and 4 of the WVS.

variables are not specifically tied to the office or workplace environment. Moreover, attitudes that are associated with grid (strong reciprocity) and group (altruism) may be operationalized to make predictions about behavior across an extremely wide range of circumstances.

At another level, more could be done to show the predictive value of grid and group–based indicators, using individual and collective behavior in experimental settings as dependent variables as well as collective political and economic outcomes as shown in comparative data sets. Indeed, we recently demonstrated that grid and group—using the WVS items and methodological approach described in this article—can successfully predict behavior in laboratory computer–mediated versions of standard experimental game theoretical setups such as the voluntary contribution mechanism (iterated *n*-person prisoner's dilemma, with or without the possibility of paying to penalize those who fail to contribute), the ultimatum game (deciding shares for a fixed amount of money, with the possibility of rejection), and the trust game (lending money to another individual where there is no compulsion for the individual to return it) (Chai, Hampton, Dorj, & Liu, 2009). Among other findings, high-group individuals are predicted and have been shown to share money more readily than others because of the tendency to identify with others, while high-grid individuals are predicted and have been shown to be willing to punish actions by others that violate norms of reciprocity, even if such punishment is costly to themselves and generates no direct benefits.

Furthermore, it may be fruitful to integrate grid and group with concepts from other cultural taxonomies as well as corresponding affective, cognitive, and motivational properties of individuals as specified by other disciplinary frameworks (e.g., from psychological and economic theory). That is, it might be possible to predict and explain the consequences of different combinations of attitudes, values, and beliefs across a wide array of disciplines. For example, perhaps grid-group concepts could be applied to the formal properties of risk avoidance and delayed gratification, constructs that are formalized and ubiquitous in microeconomic modeling but are typically

viewed as uniform, with little attention paid to the possibility of cross-cultural variability. Such a methodology could investigate whether there are similarities or differences across individuals and cultures in terms of these constructs (e.g., risk avoidance), which may help provide a more nuanced understanding of the factors and forces that shape their expression; such scholarship would seem particularly salient given the global economic meltdown of 2008 and 2009, which was plausibly determined, in no small part, by interactions among these individual and cultural variables.

Finally, the grid-group framework may be helpful in predicting between-group interaction and thus inform interventions to attenuate tensions or facilitate cooperation. That is because intergroup conflict is not simply a matter of misperception or misunderstanding but is based in real differences between groups in terms of important life values, access to resources, social power, or other significant incompatibilities (Deutsch, Coleman, & Marcus, 2000). Thus, over the long term, this approach could help societies develop a kind of metalevel awareness of the range of their characteristic belief/value positions along grid-group dimensions and how beliefs and values within one's own society are, in part, determined by particular historical forces and logics that may not be shared by other societies. Awareness of the values relating to grid and group can provide tools to creatively deal with differences among diverse groups. Recognition of and respect of value differences among distinct identity groups can help integrate seemingly conflicting points of view.

In summary, grid-group theory provides a powerful framework for exploring and explaining the nature, behavior, and dynamics of different countries and cultures. Taken together, these two dimensions can help make sense of why individual members of a culture tend to believe and value that which they do while simultaneously tracking the sum total of such beliefs and values—at the level of culture—as they shift over time. Conceptual and methodological approaches such as grid-group theory not only help illuminate the fundamental cultural parameters that shape and organize the beliefs and values we hold to be true but also provide an opportunity to bridge disciplines and methodologies and improve the quality of cross-cultural analysis. Tighter linkages between rival sets of beliefs and values and the distinctive institutions that they perpetuate and construct create clearer indices of culture. These indices contribute, in turn, to propositions that are more empirically testable, which is necessary if we are to understand better how "human nature" manifests and is organized at a social and cultural level.

NOTES

1. Available online at http://hdr.undp.org/en/statistics (accessed July 30, 2009).
2. Available online at http://earthtrends.wri.org/text/environmental-governance/variable-507.html (accessed July 30, 2009).
3. We did not consider waves 1 and 2 because the volumes of questions in waves 1 and 2 are much smaller than waves 3 and 4.
4. The 148 questions include only the value questions and did not include the identification questions.
5. For example, for grid questions 8 to 10, people have 10 choices to represent their relative opinion, but for the religion importance question, the WVS had only two choices (yes or no).

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